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☒ **Individual Author** ☐ **Collaborative (Group) Author**

Author Information

First and Middle Name (given name, e.g. John William)*

Yun-Cheng

Phil

Last Name (surname, e.g. Smith)*

Tsai

Cairns

Suffix

None

Email*

pecu610@gmail.com

pcairns@ehealthinnovation.org

Twitter Handle:

None

fnord

Role

Model Design and Experimental Results with Training and Validation

Technical Analyst

ORCID iD

0000-0002-3830-3272

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Author Information

?First and Middle Name (given name, e.g. John William)*

Yi-Lin

Phil

?Last Name (surname, e.g. Smith)*

Wu

Cairns

?Suffix

None

?Email*

lynnewu727@gmail.com

pcairns@ehealthinnovation.org

?Twitter Handle:

None

fnord

?Role

Conceptualization, Methodology, Formal Analysis, Investigation, Data Curatic

Technical Analyst



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PharmB
PharmD
PhB
PhD
PhL

new degree



PhD



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Author Information

?First and Middle Name (given name, e.g. John William)*

Yung-Chen

Phil

?Last Name (surname, e.g. Smith)*

Yu

Cairns

?Suffix

None

?Email*



fnord

?Role

Conceptualization, Investigation, Data Curation, Writing-Review and Editing

Technical Analyst

?ORCID iD

0000-0003-1857-9762

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?DegreesA
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AAAS new degree

MS



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?Individual Author **?Collaborative (Group) Author****Author Information****?First and Middle Name (given name, e.g. John William)***

Yi-Wen

Phil



None

Email*

maichann921@gmail.com

pcairns@ehealthinnovation.org

Twitter Handle:

None

fnord

Role

Validation, Investigation, Writing-Review and Editing

Technical Analyst

ORCID iD

0000-0002-9935-7100

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PharmB

PharmD

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new degree

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Affiliation Information

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**?First and Middle Name (given name, e.g. John William)***

Yi-Wen

Phil

?Last Name (surname, e.g. Smith)*

Chen

Cairns

?Suffix

None

?Email*

owenchen@jubo.health

pcairns@ehealthinnovation.org

?Twitter Handle:

None

fnord

?Role

Software, Formal Analysis, Resources, Data Curation

Technical Analyst

?ORCID iD

0000-0002-3585-1767

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Author Information

? First and Middle Name (given name, e.g. John William)*

Yi-Ching

Phil

? Last Name (surname, e.g. Smith)*

Yang

Cairns

? Suffix

None

? Email*

yiching@mail.ncku.edu.tw

pcairns@ehealthinnovation.org

? Twitter Handle:

None

fnord

? Role

Resources, Writing-Review and Editing, Supervision

Technical Analyst

? ORCID iD

0000-0003-1391-8040

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Phil

?Last Name (surname, e.g. Smith)*

Cairns

?Suffix**?Email***

pcairns@ehealthinnovation.org

?Twitter Handle:

fnord

?Role

Technical Analyst

?ORCID iD

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(Optional: specific department or unit within the faculty or division)
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Address*

Address Line 1 (mandatory)*.
Example: TGH/RFE Bldg, 4th Fl.

Address Line 2 (optional).
Example: 190 Elizabeth St

City*

Example: Bethesda

Postal/Zip Code

Example: M5G 2C4

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Example: Canada

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Example: 1-416-340-3200

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Issue or Intervention in Demographic/Disease/Condition: Method/Study Design

Development and Temporal External Validation of a Parsimonious, Interpretable Machine Learning Model for Predicting 6-Month Mortality in Long-Term Care Facilities: A Retrospective Cohort Study

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Abstract Type: ☒ Structured ☐ Unstructured



To develop and temporally externally validate an interpretable machine learning model for predicting 6-month all-cause mortality among older adults newly admitted to LTCFs in Taiwan using routinely collected admission data.



Methods*

We conducted a retrospective cohort study using the JUBO Long-Term Care Database, a nationwide private administrative regi care facilities in Taiwan). We included residents aged ≥ 65 years with first-time LTCF admission and prespecified nonoverla (January 1, 2020–December 31, 2023; $n = 23,901$) and external validation (January 1–December 31, 2024; $n = 6216$). The outco We compared a nonlinear ensemble model (HybridXGBRF) with seven other algorithms, including tree-based and linear benchmar (accuracy, precision, recall, and F1), and calibration (Brier score and calibration plots) were assessed. Model interpreta Explanations (SHAP).



Results*

In the development cohort, 5272/23,901 (22.1%) residents died within six months; in the 2024 validation cohort, 1781/6216 HybridXGBRF model achieved the highest AUROC (0.875, 95% CI 0.862–0.889) and the lowest Brier score (0.109). In the tempor maintained strong discrimination (AUROC 0.878, 95% CI 0.866–0.889), with an accuracy of 0.851 and an F1 score of 0.572. Ca the predicted and observed risks across most probability ranges in both cohorts, with a mild divergence at higher predicte hospitalizations within six months, activities of daily living impairment, and weight loss as the most influential predict age strata (0.88–0.89), and maintained high discrimination in the subgroup of residents with improving activities of daily mortality risk is often underestimated by conventional clinical assessments.



Conclusions*

An interpretable machine learning model using routinely collected admission data provide accurate mortality predictions without requiring hospital-based electronic health record data linkages, and generalized well under temporal validation in Taiwan. This practical tool can serve as a decision-support system to prompt early palliative care discussions and enhance person-centered care planning in LTCF.



Trial Registration

Not applicable



Indexing

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long-term care facilities; mortality prediction; machine learning; artificial intelligence; Taiwan

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Collaborators

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